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SUPERSEDING
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MILITARY SPECIFICATION

POTATO PATTIES, PRECOOKED, DEHYDRATED, FOR MEAL, READY-TO-EAT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers dehydrated precooked potato patties in flexible bags for use as a component of the meal, ready-to-eat, individual.

2. APPLICABLE DOCUMENTS

* 2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein:

SPECIFICATIONS

FEDERAL

L-P-378	-	Plastic Sheet and Strip, Thin Gauge, Polyolefin
EE-S-631	-	Spices, Ground and Whole
QQ-A-1876	-	Aluminum Foil

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research and Development Laboratories, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8915

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PPP-B-636 - Boxes, Shipping, Fiberboard
PPP-B-26 - Bags, Plastic, Polyethylene
PPP-C-843 - Cushioning Material, Cellulosic
PPP-C-1752 - Cushioning Material, Packaging, Unicellular Polyethylene Foam, Flexible
PPP-C-1797 - Cushioning Material, Resilient, Low Density, Unicellular, Polypropylene Foam

STANDARDS

FEDERAL

FED-STD-595 - Colors

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Copies of documents required by manufacturers in connection with specific acquisition functions, should be obtained from the contracting activity or as directed by the contracting officer.)

OTHER GOVERNMENT DOCUMENTS

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder (21 CFR Parts 1-199)

(Copies may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

ENVIRONMENTAL PROTECTION AGENCY (EPA)

National Interim Primary Drinking Water Regulations

(Application for copies should be addressed to the Office of Drinking Water WH550, Environmental Protection Agency, Washington, DC 20460.)

2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

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USDA COLOR STANDARDS FOR FROZEN FRENCH FRIED POTATOES

(Application for copies should be addressed to Munsell Color Company, Inc., 2441 North Calvert Street, Baltimore, MD 21218.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 882 - Tensile Properties of Thin Plastic Sheeting

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

ASSOCIATION OF OFFICIAL ANALYTICAL CHEMISTS

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, 1111 No. 19th St., Arlington, VA 22209.)

AMERICAN DEHYDRATED ONION AND GARLIC ASSOCIATION (ADOGA)

Official Standards and Methods of the American Dehydrated Onion and Garlic Association (ADOGA) for Dehydrated Onion and Garlic Products

(Application for copies should be addressed to the American Dehydrated Onion and Garlic Association, 251 Kearney Street, No. 410, San Francisco, CA 94108.)

NATIONAL ACADEMY OF SCIENCES

Food Chemicals Codex

(Application for copies should be addressed to the Office of the Director, National Academy Press, 2101 Constitution Avenue, N.W., Washington, DC 20418.)

AMERICAN OIL CHEMISTS SOCIETY

Official and Tentative Methods of the American Oil Chemists Society

(Application for copies should be addressed to the American Oil Chemists Society, 508 South Sixth Street, Champaign, IL 61820.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.4, 6.1, and 6.3).

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent or insect infestation, extraneous material, off-odors, off-flavors, and off-colors.

3.2.1 Potatoes. The potatoes shall be fresh potatoes of one variety, such as Russet Burbank, Norgold Russet, Kennebec, Green Mountain, Cobbler, Red McClure, Sebago or Katahdin.

3.2.2 Frying fat and shortening. Frying fat and shortening shall be a hydrogenated vegetable fat or oil having good flavor and odor. It shall have a stability of not less than 40 hours as determined by the active oxygen method in Method Cd 12-57 and a smoke point of not less than 400°F as determined by Method Cc 9a-48 of the Official and Tentative Methods of the American Oil Chemists Society. Antioxidants permitted by the Field Operations Division of the USDA may be added in amounts permitted by regulations. Addition of antioxidants is required when patties will be held more than 30 days from frying to start of dehydration (see 3.3.4).

3.2.3 Corn meal. The corn meal shall be a high quality degerminated yellow, commercially available product.

3.2.4 Onions dehydrated. The onions shall be dehydrated, minced onions or onion powder "fancy grade" of the Official Standards and Methods of the American Dehydrated Onion and Garlic Association (ADOGA) for Dehydrated Onion and Garlic Products.

3.2.5 Monosodium glutamate (MSG). The MSG shall be free-flowing crystalline powder conforming to Food Chemicals Codex.

3.2.6 Sodium bisulfite. Sodium bisulfite shall conform to the requirements of the Food Chemicals Codex.

3.2.7 Salt. Salt shall be noniodized, clean, white, refined sodium chloride with or without anticaking agents.

3.2.8 Pepper. The pepper shall be ground white pepper complying with type II of EE-S-631.

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3.2.9 Corn meal substitutes. Corn flour, potato flakes, potato starch and potato granules shall be of good commercial grade and suitable for use in foods.

3.2.10 Optional ingredients. Dextrose and disodium dihydrogen pyrophosphate of good commercial quality and suitable for use in foods may be used in conjunction with corn meal substitutes.

* 3.2.11 Water. Water used for ice making, formulation, and washing shall conform to the National Interim Primary Drinking Water Regulations.

3.3 Preparation and processing.

3.3.1 Potato preparation. The potatoes shall be peeled and shredded into 1/4 by 1/4 inch shred squares (in cross section), or other dimensions suitable for meeting end product requirements. If necessary to meet color requirements, the peeled potatoes shall be maintained in a solution of sodium bisulfite (2 grams sodium bisulfite per gallon of water) prior to blanching. The shreds shall be blanched or cooked sufficiently to assure acceptable end item characteristics (see 3.4).

3.3.2 Patty preparation. The patties shall be formulated as follows:

Ingredient	Percent
Potato 1/	69.24
Water, cold 2/	21.00
Corn meal 3/	7.00
Salt	1.10
Shortening	1.00
Onions, dehydrated	0.50
Monosodium glutamate	0.10
White pepper	0.06

1/ Alternatively, an additional percentage of potato, above the required 69.24 percent, may be used in lieu of an equal percentage of water for formulation adjustment.

2/ If necessary, the amount of water may be varied to facilitate the manufacturing process.

3/ Corn flour, potato flakes, potato starch, potato granules, dextrose, disodium dihydrogen pyrophosphate may be substituted provided the finished product meets the requirements in 3.4. The formulation shall be adjusted accordingly.

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The shortening, corn meal or alternates, seasonings, and water (cold) shall be added together, thoroughly mixed, heated until the corn meal gelatinizes, and the total mixture reaches a temperature of 180°F. The blanched potato shreds shall be added to the mixture and shall be mixed into the mass long enough to assure uniform distribution of the shreds without mushing. When corn flour, potato flakes, potato starch, dextrose or disodium dihydrogen pyrophosphate are substituted, the cooking provisions need not apply. The potato patty mixture shall be molded into patties of sufficient size, so that after deep frying, the patties will conform to the dimensions and weight requirements of 3.4.

3.3.3 Patty frying. The formed potato patties shall be fried in deep fat at a temperature of 360° to 375°F, or processed as necessary to meet end item requirements. The cooked patties shall contain no blemishes, discoloration, or similar defects affecting end product edibility or characteristic appearance above the levels listed below for each category. Very light discolorations, light or dark brown surface or internal discolorations smaller in size than that of category A, and carbon specks, usually 1 mm or less in size, shall be disregarded unless, in total, they more than slightly affect the overall appearance. After frying, the product shall be drained of excess fat.

Category	Light brown surface or internal discoloration	Dark brown surface or internal discoloration	Maximum allowable average No. of defects per patty (see 4.5.3)
A-Slight	Equal to or larger than the area of a circle or volume of a sphere of 6 mm diameter but less than the area of a circle or volume of a sphere of 8 mm diameter.	Equal to or larger than the area of a circle or volume of a sphere of 3 mm diameter but less than the area of a circle or volume of a sphere of 6 mm diameter.	3
B-Moderate	Equal to or larger than the area of a circle or volume of a sphere of 8 mm diameter.	Equal to or larger than the area of a circle or volume of a sphere of 6 mm diameter.	1
	Any internal/external condition which is uncharacteristic of the product appearance.		
C-Serious	Inedible particles or offensive, uncharacteristic appearance.		1/2

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3.3.4 Patty freezing. The patties shall be placed in a blast-freezer within 4 hours after completion of cooking and frozen to 0°F or lower within 6 hours after placement in the freezer. The temperature of the frozen patties shall not be allowed to exceed 5°F at anytime during handling and storage prior to dehydration. Maximum time from frying to start of dehydration shall be 30 days; 120 days if antioxidants are incorporated as stated in 3.2.2.

3.3.5 Freeze dehydration. The patties shall be freeze-dehydrated at an absolute pressure not to exceed 1.5 millimeters of mercury and a patty temperature, as indicated by suitable instruments, of not more than 150°F. If the platen temperature is maintained at 155°F or below, the patty temperature may be disregarded. After dehydration has been completed, the pressure shall be equalized to atmospheric level with nitrogen and the patties shall be packaged without delay as specified in 5.1. In no case shall more than 8 hours elapse between the time the chamber is opened and the time the patties are completely packaged. During the interim, the patties shall be adequately protected from oxygen and moisture by either holding under a nitrogen atmosphere with 2.0 percent or less oxygen, or under a vacuum of at least 27 inches for the entire period. If vacuum is used, it shall be broken with nitrogen.

3.4 Finished product. The finished product shall comply with the following requirements:

a. There shall be no foreign material, such as, but not limited to dirt, insect, insect part, hair, wood, paper, paint, glass or metal.

b. There shall be no foreign odor or flavor, such as, but not limited to burnt, scorched, stale, sour, rancid, musty, or moldy.

c. There shall be no evidence of incomplete dehydration such as damp or soggy areas.

d. There shall be no evidence of faulty dehydration such as glazed surface areas measuring more than 0.5 inch in any dimension or dark colored cores of any dimension.

e. The overall color of the potato patty shall range from No. 2 to No. 3 with no patty lighter than No. 1 or darker than No. 4 in accordance with the USDA Color Standards for Frozen French Fried Potatoes.

f. The product shall contain distinct pieces of potato.

g. The product shall conform to the following dimensions:

Length	Width	Thickness
3.5 to 4.25 inches	1.9 to 2.9 inches	0.5 to 0.8 inches

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h. The dehydrated patty average weight shall be 27 grams minimum and no individual dehydrated patty shall weigh less than 25 grams.

i. The patties shall not be crushed nor broken into more than two pieces. Occasional pieces of potatoes which have separated from the patty shall not be considered as broken pieces.

j. Moisture content of the dehydrated product shall not exceed 3.0 percent.

k. The product shall rehydrate when 3 fluid ounces of hot water (180° to 210°F) are added to an opened bag of product and the product is immersed completely and allowed to stand for 5 minutes. Complete immersion of the patty shall be effected by adjusting the bag.

3.4.1 Palatability. The product shall be equal to or better than the approved preproduction sample (see 6.1) in palatability and overall appearance.

3.5 Plant qualification. The product shall be prepared, processed and packaged under sanitary conditions in accordance with good manufacturing practice. The product shall be processed in establishments meeting the sanitary requirements of the applicable Government inspection agency.

3.6 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all requirements of this document. The contractor shall assure product compliance prior to submitting the product to the USDA for any inspection.

4.2 Inspection and acceptance service. Product acceptability shall be determined by the USDA. The USDA will determine the degree of acceptance service necessary to assure compliance with the requirements of this document. The cost of grading and acceptance services performed by the USDA involving inspection, official documentation, and related services shall be borne by the contractor.

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4.3 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

* 4.4 First article inspection. When a first article is required (see 6.1), it shall be inspected in accordance with the quality assurance provisions of this document and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this document or any appearance or palatability failure shall be cause for rejection of the first article.

4.5 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.5.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.5.1.1 Ingredient examination. Conformance of ingredients to identity and condition requirements shall be determined by examination of pertinent labels, markings, U.S. Grade certificates, invoices, certificates of analysis, or other valid documents. In addition each ingredient shall be examined organoleptically as necessary to determine conformance to the condition requirements. Any nonconforance to an identity, condition, or other requirement shall be cause for rejection of the ingredient lot or of any involved product.

4.5.1.2 Laminated bag material certification. Material listed below shall be accepted on the basis of a contractor's certification of conformance to the indicated requirements.

Material requirement	Requirement paragraph	Test procedure
Ionomer or polyethylene film thickness	5.1.1	As specified L-P-378, except that a machinist's micrometer may be used provided that its graduations and accuracy conform to the requirement of L-P-378.
Polyester film thickness	5.1.1	As above
Aluminum foil thickness	5.1.1	As specified in QQ-A-1876
Laminated material construction	5.1.1	Laboratory evaluation
Color of laminated material	5.1.1	Visual evaluation

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* 4.5.1.3 Polyethylene bag examination. The unfilled bags shall be examined for the defects listed in table I. The lot size shall be expressed in bags. The sample unit shall be one unfilled bag. The inspection level shall be S-4 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 0.65.

TABLE I. Unfilled polyethylene bag defects

Category	Defect
Major	
151	Not size specified
152	Individual bags that stick together or that tear when separated.
153	Not material specified
154	Tear, hole or open seal
155	Not clean

* 4.5.1.4 Preformed laminated bag examination (when applicable). The unfilled bags shall be examined for the defects listed in table II. The lot size shall be expressed in bags. The sample unit shall be one unfilled bag. The inspection level shall be S-4 and the AQL, expressed in terms of defects per hundred units, shall be 0.65 for major defects and 4.0 for minor defects.

TABLE II. Unfilled preformed laminated bag defects

Category		Defect
Major	Minor	
151		Not size specified
152		Heat seal dimensions on three edges not as specified
153		Tear, hole or open seal
154		Not clean
155		Color not as specified
156		Required labeling missing, incorrect, or illegible

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TABLE II. Unfilled preformed laminated bag defects (cont'd)

Category		Defect
Major	Minor	
157		Not material specified
158		Individual bags that stick together or that tear when separated
	201	Tear notch missing
	202	Tear notch not located as specified
	203	Depth of tear notch not as specified.

4.5.1.5 Preformed laminated bag seal strength testing (when applicable). The unfilled bags shall be tested for seal strength in accordance with Method A or B of ASTM D 882 except that testing speed may be 10 or 12 inches per minute. The lot size shall be expressed in bags. The sample size shall be the number of bags indicated by inspection level S-1. Three adjacent specimens shall be cut from each of the three sealed sides of each bag in the sample. The results shall be reported to the nearest 0.1 pound. The average seal strength of each seal shall be calculated by averaging the strengths of the three test specimens cut from that seam. Any test specimen failing to meet the individual test specimen seal strength requirement specified in 5.1.1 or any seal failing to meet the average seal strength requirement specified in 5.1.1 shall be cause for rejection of the lot.

4.5.2 In-process examination. In-process examination shall be performed to determine conformance to preparation, processing, holding, bag filling, and bag closing requirements. Any nonconformance revealed by actual examination or by review of records of time, temperature, and formulation or other valid documents shall be cause for rejection of the involved product.

4.5.3 In-process examination for frying defects. The patties, after frying and prior to dehydration, shall be examined for the edibility and appearance defects indicated in 3.3.3 and the defects found shall be categorized as specified herein. The sample size shall be the number of patties indicated by inspection level S-2. The number of defects in each category shall be divided by the number of patties in the sample. If the average number of defects per patty for any category exceeds that allowed in 3.3.3, the lot shall be rejected.

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4.5.4 Patty weight inspection. The weight of the dehydrated patties shall be determined by weighing each sample unit on a suitable scale tared with an average weight of representative empty laminated bags and polyethylene bags. Any individual weight of less than 25 grams shall be scored as a minor defect. The lot size shall be expressed in bags. The sample unit shall be one filled and sealed bag. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 2.5. In addition, the lot shall be rejected if the sample average patty weight is less than 27 grams.

* 4.5.5 Laminated bag vacuum examination. The filled and sealed bags shall be visually examined for a proper vacuum level not less than 48 hours after filling and sealing. The sealed bag shall continue to exhibit tight adherence to the surface contours of the contents when a pulling force is applied at the center of each side seal. This force shall be applied by holding each side seal between thumb and forefinger of each hand, simultaneously exerting a slight pull with both hands. Any evidence of loss of vacuum shall be considered a major defect. The lot size shall be expressed in bags. The sample unit shall be one filled and sealed bag. The inspection level shall be I (but not less than 200 sample units) and the AQL, expressed in terms of defects per hundred units, shall be 0.65.

* 4.5.6 Filled and sealed laminated and polyethylene bag examination. The filled and sealed laminated and polyethylene bags shall be examined for the defects listed in table III. The lot size shall be expressed in bags. The sample unit shall be one filled and sealed bag. The inspection level shall be I (but not less than 200 sample units) and the AQL, expressed in terms of defects per hundred units, shall be 0.65 for major defects and 2.5 for minor defects.

TABLE III. Filled and sealed laminated and polyethylene bag examination 1/

Category		Defect
Major	Minor	
		Laminated bag
151		Not formed as specified 2/
152		Not clean 2/
153		Tear, hole, or open seal 2/
154		Closure seal not produced by specified method

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TABLE III. Filled and sealed laminated and polyethylene bag examination 1/ (cont'd)

Category		Defect
Major	Minor	
155		Evidence of delamination 3/
156		Closure seal width not as specified
157		Closure or top seal extends into or below tear notch 4/
158		Bag has foreign odor
159		Required labeling missing, incorrect, illegible, or that smudges
160		Color not as specified 2/
161		Tear notch or serrations missing
162		Tear notch or serrations not located as specified
163		Depth of tear notch or serrations not as specified
164		Inner bag caught in closure seal making removal of contents impossible without rendering the laminated bag unfit for rehydration of the contents.
165		Presence of flex fractures in the aluminum foil 2/ 5/
	201	Inner bag caught in closure seal, but contents can be removed without making the laminated bag unfit for rehydration of the contents
		Polyethylene bag
	202	Open end of bag not closed by folding end back over the patty

1/ Any evidence of insect or rodent infestation shall be cause for rejection of the lot.

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2/ Applicable to form-fill-seal bags only.

3/ Delamination shall be scored as a defect except delamination of outer ply when located in the seal area 1/16 inch or further from food product edge of seal. Bags exhibiting this type of delamination shall be tested by manually flexing the delaminated area 10 times. The area of delamination shall be held between the thumb and forefinger of each hand with both thumbs and forefingers touching each other. The delaminated area shall then be rapidly flexed by rotating both hands in alternating clockwise - counterclockwise directions. Care shall be exercised when flexing delaminated area near the tear notches to avoid tearing the bag material. After flexing, the separated outer ply shall be grasped between the thumb and forefinger and gently lifted toward the food product edge of the seal. If the separated area is too small to be held between thumb and forefinger, a number two stylus shall be inserted into the delaminated area and a gentle lifting force applied against the outer ply. If separation of the outer ply can be made to extend to less than 1/16 inch from the product edge of the seal with no discernible resistance to the gentle lifting, the bag shall be rejected.

4/ Applicable to preformed bag only.

5/ To test for flex fractures the inside surface of the tray-shaped body shall be placed over a light source and the outside surface observed for the passage of light. Observation of light through the bag shall be evidence of the presence of flex fractures in the aluminum foil.

4.5.7 Product examination. The finished product shall be examined for the defects listed in table IV. The lot size shall be expressed in bags. The sample unit shall be the contents of one filled and sealed bag. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 1.5 for major defects and 6.5 for minor defects.

TABLE IV. Product examination 1/ 2/

Category		Defect
Major	Minor	
		Applicable to dehydrated product:
151		Evidence of incomplete dehydration such as damp or soggy areas
152		Presence of glazed surface areas measuring more then 0.5 inch in any dimension

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TABLE IV. Product examination 1/ 2/ (cont'd)

Category		Defect
Major	Minor	
153		Color not as specified (see 3.4e)
154		Does not contain distinct pieces of potato
155		Patty is broken or crushed 3/
	201	Length of patty less than 3.5 inches or more than 4.25 inches
	202	Width of patty less than 1.9 inches or more than 2.9 inches
	203	Thickness of patty less than 0.5 inch or more than 0.8 inch
156		Applicable to rehydrated product 4/ Product does not rehydrate within 5 minutes (presence of dry particles when patty is broken into pieces)

1/ The presence of any foreign material (e.g. glass, dirt, insects, insect parts, wood, metal, etc.), foreign odor off-flavor (e.g. burnt, scorched, moldy, rancid, sour, stale, etc.) or foreign color shall be cause for rejection of the lot.

2/ Product not equal to or better than the approved preproduction sample (see 6.1) in palatability or overall appearance shall be cause for rejection of the lot. (This comparison shall be performed only when deemed necessary by an AMS agent.)

3/ A broken patty is one that is broken into three or more distinct pieces or which has more than 1/8 of its volume missing. A crushed block is one in which 1/8 or more of the volume of the block has been reduced to crumbs or powder.

4/ Rehydrate by adding 3 fluid ounces of hot water (180° to 210°F) to the opened bag so as to completely cover the patty and allow to stand 5 minutes.

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* 4.5.8 Laminated bag closure seal testing. The filled and sealed bags shall be tested for closure seal strength in accordance with method A or B of ASTM D 882 except that the testing speed may be 10 or 12 inches per minute. The lot size shall be expressed in bags. The sample unit shall be one filled and sealed bag. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 1.5. For the preformed bag, three adjacent specimens, 1/2 inch or 1-inch wide, shall be cut from the top closure seam of each bag in the sample. For the alternate form-fill-seal bag which, has all four sides closed simultaneously, three adjacent specimens, 1/2 inch or 1-inch wide, shall be cut from each of the four sides of each bag. The results shall be reported to the nearest 0.1 pound per linear inch. For the preformed bag, the average seal strength of the closure seal shall be calculated by averaging the test results of the three test specimens cut from that seal. For the form-fill-seal bag the average seal strength of each closure seal shall be calculated by separately averaging the test results of the three specimens cut from each side seal. Any test specimen failing to meet the individual test specimen seal strength requirement specified in 5.1.1 or any closure seal failing to meet the average seal strength requirement specified in 5.1.1 shall be scored a defect.

* 4.5.9 Moisture content testing. The finished product shall be prepared and tested for moisture content in accordance with the Moisture in Dried Fruits Method in the chapter on Fruits and Fruit Products of the Association of Official Analytical Chemists. The results shall be reported to the nearest 0.1 percent. Any result not conforming to the requirement in 3.4 shall be considered a defective. The lot size shall be expressed in bags. The sample unit shall be one filled and sealed laminated bag. The inspection level shall be S-2 and the AQL, expressed in terms of percent defective, should be 1.5.

4.5.10 Shipping container examination. Examination of the filled and closed shipping containers shall be in accordance with the examination criteria contained in the appendix of PPP-B-636, with the following defects added to the table: Major: (1) Marking missing, incorrect or illegible; (2) Cushioning material missing or not as specified; Minor: (1) Number of bags more than 5 percent over or under the number specified; (2) Bags not packed flat in layers.

5. PACKAGING

5.1 Preservation. Preservation shall be level A.

5.1.1 Level A. Each patty shall be placed in a minimum 0.003 inch thick, food grade, polyethylene bag fabricated in accordance with type II, grade A, style 1 of PPP-B-26. The polyethylene bag shall be of such size as to

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facilitate bag filling, allow for bag closure as specified, and facilitate subsequent filling and closure of laminated bag with no obstruction from excess material. Each patty shall be placed into the polyethylene bag with one complete edge against the bag bottom. The open end of the bag shall be folded against the nearest patty edge and extend over one of the largest adjacent patty surfaces a minimum of 1-1/2 inches and a maximum of the whole length or width of that surface as applicable. The patty in the polyethylene bag shall be unit packed in a laminated bag. The bag material shall be a 3 ply laminate consisting of 0.002 inch thick ionomer or polyethylene film laminated or extrusion coated to 0.00035 inch thick aluminum foil which is then laminated to 0.0005 inch thick polyester. The three plies shall be laminated so that the aluminum foil is between the other two layers. The bag shall be formed with the polyester on the exterior of the bag. The complete exterior surface of the bag shall be uniformly colored in the range of green 34079 through 34087 or 24052 through 24087 or brown 30045 through 30118 (excluding 30109) or 10045 of FED-STD-595. The laminated bag shall be formed by heat sealing to the size and design configuration as shown in figure 1. The laminated bag may be preformed (sealed on three sides) then filled and sealed at the open end, or, the laminated bag may be formed, filled and seals simultaneously applied on all four sides. The end of the laminated preformed bag may be provided with a 1/4 $\pm$ 1/16 inch extended or fold-over lip to facilitate opening and filling. Tear notches as indicated in figure 1 shall be provided on preformed bags or bags that are formed, filled and simultaneously sealed on all four sides. Alternatively, if the bag has serrated edges, the serrations may be used as tear notches provided that the serrations are sharp (no plastic tailings exist) and the serration depth and the minimum seal width at the serrated edges are in accordance with the dimensions shown in figure 1. The filled bag shall be closed under a vacuum of not less than 25 inches of mercury (see 4.5.5). Closure of the preformed bag shall be effected with a continuous heat seal not less than 1/4 inch wide. If thermal impulse or combination (heated curved bar with thermal impulse) sealing is used, any seal width from 1/8 to 7/16 inch will be acceptable. The closure or top seal location shall be as shown in figure 1. Closure of the form fill seal bag shall be accomplished by application of all four seals simultaneously. All four seals shall be 3/8 $\pm$ 1/8 inch wide. The material shall show no evidence of delamination or degradation when heat sealed or fabricated into bags and shall not transfer any foreign odor or flavor to the unit packed product. The average seal strength of each seal shall be not less than 6 pounds per linear inch, and no individual test specimen shall be less than 5 pounds per linear inch when tested as specified in 4.5.1.5 and 4.5.8.

5.2 Packing. Packing shall be level C.

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* 5.2.1 Level C, for shipment to ration assembly. Not more than 325 patties preserved as specified in 5.1.1 shall be packed flat in a fiberboard box conforming to style RSC-L, class domestic, variety SW, grade 175 of PPP-B-636. Cushioning material not less than 1/8 inch thick, conforming to type II, class B of PPP-C-843 or cushioning material not less than 1/16 inch thick conforming to PPP-C-1797 or type 7, class 1 of PPP-C-1752 shall be placed between each of the layers and in the bottom and top of the box to minimize movement of the individual packages.

5.3 Labeling and marking.

5.3.1 Unit packs. Each unit pack (bag) shall be clearly printed with permanent ink, in large clear black letters with the following information:

POTATO PATTY

EAT DRY OR RECONSTITUTE IN HOT OR COLD WATER.
TO RECONSTITUTE, COVER PATTY WITH WATER.
LET SOAK UNTIL SOFT (2-5 minutes).
DRAIN.
Name and address of producer.

5.3.2 Shipping containers. Shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. When a first article is required (see 3.1, 4.4 and 6.3).
c. Provisions for approved preproduction samples (see 3.4 and 6.3).

6.2 Award of contract. Award of contract for the product specified in this document will be limited to plants known to maintain the required sanitation conditions of 3.5.

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of DAR 7-104.55. The first article should be a preproduction sample. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

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6.4 Changes from previous issue. The margins of this document are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Navy - SA
Air Force - 50

Preparing activity:

Army - GL

Project No. 8915-0780

Review activities:

Army - MD, TS
Navy - MC, MS
DP - SS

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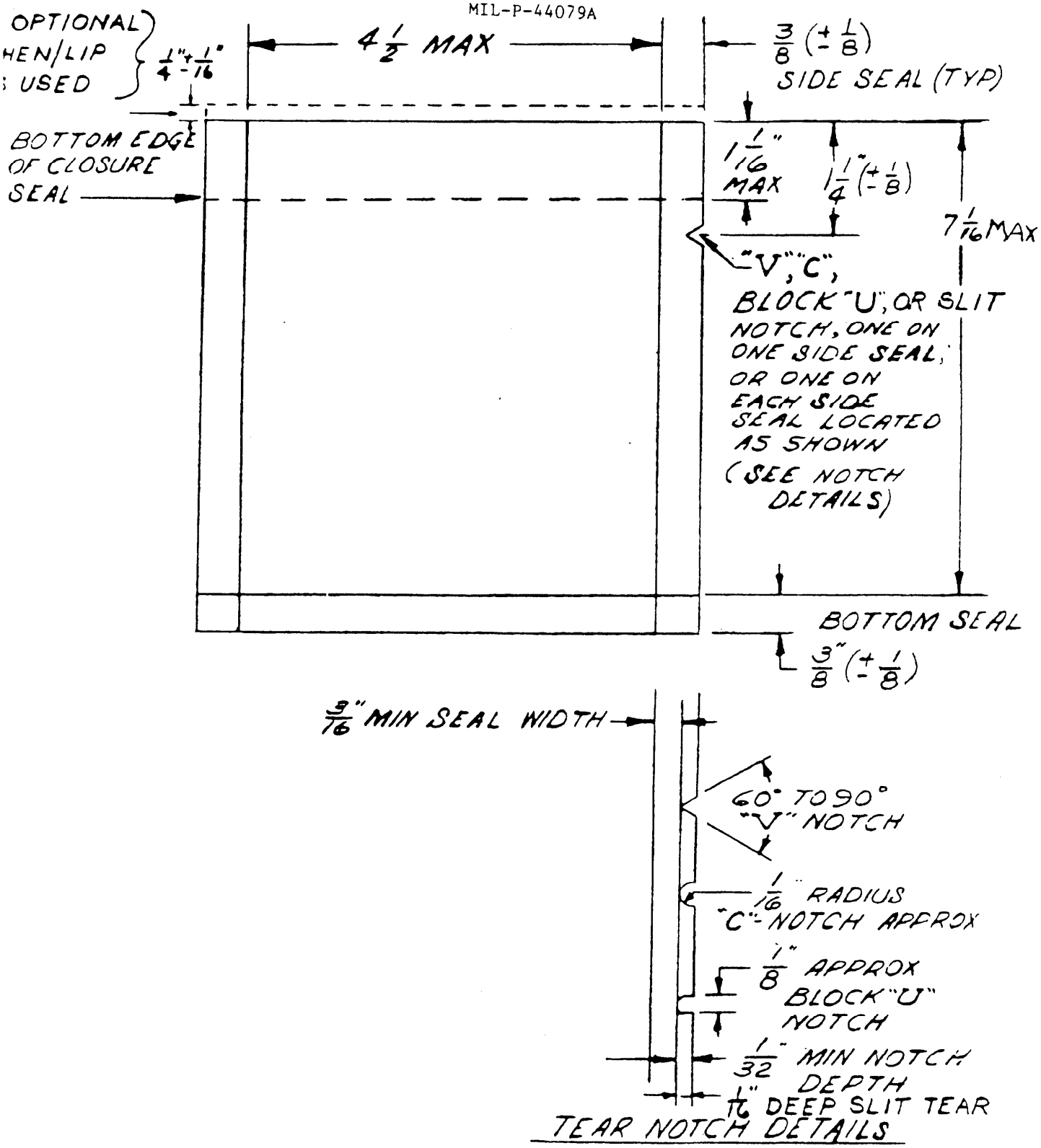


FIGURE 1

FOUR SEAL BAG

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-44079A	Potato Patties, Precooked, Dehydrated, For Meal, Ready-To-Eat

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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